

COMPANION
to

ChatGPT Teaches TikZ

Chapter 25: Externalizing TikZ Pictures (A)

Contents

25.1	What This Companion Adds	2
25.2	An Externalization Vocabulary	3
25.3	Deciding Whether to Externalize	4
25.4	The Minimal Setup	4
25.5	Keeping the Cache Separate	5
25.6	Stable Picture Names	5
25.7	Selective Participation	5
25.8	Rebuilding the Right Amount	6
25.9	Shared Definitions and Stale Output	6
25.10	A Safe Project Layout	6
25.11	Worked Project: Two Named Pictures	7
25.12	A Production Checklist	8
25.13	Debugging Workshop	8
25.14	Exercises	9
25.15	Solutions	10
25.16	ChatGPT Workshop	11
25.17	Final Checklist	11

25.1 What This Companion Adds

Chapter 25 introduced externalization as a production technique for large TikZ projects. This Companion develops a cautious workflow: debug pictures normally, name stable figures, isolate generated files, rebuild the smallest necessary part of the cache, and verify a clean production build.

Design

Externalization changes how a finished picture is built, not how it is drawn. Make the figure correct before adding production machinery.

25.2 An Externalization Vocabulary

Command or option	Purpose
<code>\usetikzlibrary{external}</code>	Loads the externalization library.
<code>\tikzexternalize</code>	Activates externalization.
<code>prefix=</code>	Selects a directory for generated files.
<code>\tikzsetnextfilename</code>	Names the next externalized picture.
<code>external/remake next</code>	Rebuilds only the next picture.
<code>external/force remake</code>	Requests a broad rebuild.
<code>external/export</code>	Skips externalizing the next picture.
<code>next=false</code>	
<code>\tikzexternaldisable</code>	Temporarily disables externalization.
<code>-shell-escape</code>	Permits the auxiliary compilation process.

25.3 Deciding Whether to Externalize

Externalization is valuable when unchanged complex figures dominate compilation time. It adds little value to a short document containing a few simple pictures.

Measure first. Develop normally while figures change frequently. Introduce the cache when repeated rebuilding becomes an actual obstacle.

25.4 The Minimal Setup

In the preamble, load the library and activate it.

```
\usetikzlibrary{external}  
\tikzexternalize
```

The ordinary picture source remains unchanged.

```
\begin{tikzpicture}  
  \node[draw] (A) at (0,0) {$A$};  
  \node[draw] (B) at (3,0) {$B$};  
  \draw (A)--(B);  
\end{tikzpicture}
```

The usual command-line workflow permits the auxiliary compiler.

```
pdflatex -shell-escape main.tex
```

Common Mistake

The `-shell-escape` option allows external commands. Use it only with a document and package set that you trust.

25.5 Keeping the Cache Separate

```
\tikzexternalize[prefix=tikz-cache/]
```

Create the directory before compiling when the TeX environment does not create it automatically. The cache contains generated artifacts; the `.tex` source remains authoritative.

25.6 Stable Picture Names

```
\tikzsetnextfilename{triangle-median}  
\begin{tikzpicture}  
  % triangle and median  
\end{tikzpicture}
```

Names should describe mathematical content rather than page number or temporary figure order. Stable names make the cache understandable after figures move.

25.7 Selective Participation

Some pictures are too small to benefit or depend on surrounding document material.

```
\tikzset{external/export next=false}  
\begin{tikzpicture}  
  % compile this picture normally  
\end{tikzpicture}
```

For a longer region, disable and re-enable externalization explicitly.

```
\tikzexternaldisable  
% pictures compiled normally  
\tikzexternalenable
```

Use exclusions deliberately and document why they are needed.

25.8 Rebuilding the Right Amount

To rebuild only one known picture:

```
\tikzset{external/remake next}  
\tikzsetnextfilename{triangle-median}  
\begin{tikzpicture}  
  % revised picture  
\end{tikzpicture}
```

Use a broad forced rebuild when shared styles, libraries, or macros affect many cached pictures.

```
\tikzset{external/force remake}
```

Do not edit a generated PDF to fix a figure. Edit the TikZ source and regenerate the cache.

25.9 Shared Definitions and Stale Output

A picture may look unchanged to the cache even after an indirect dependency has changed. Project-wide styles, custom keys, pics, fonts, and included data can all affect the rendered result. After such changes, rebuild the affected pictures or perform a clean production rebuild.

25.10 A Safe Project Layout

Location	Contents
main.tex	The authoritative manuscript source.
TikZstyle.sty	Shared authored styles and reusable components.
figures/	Optional authored figure source files.
tikz-cache/	Generated externalized graphics and support files.

Source and generated material now have distinct roles.

25.11 Worked Project: Two Named Pictures

25.11.1 Stage 1: Configure the Preamble

```
\usetikzlibrary{external}
\tikzexternalize[prefix=tikz-cache/]
```

25.11.2 Stage 2: Name Each Picture

```
\tikzsetnextfilename{graph-path-four}
\begin{tikzpicture}
  % first finished figure
\end{tikzpicture}

\tikzsetnextfilename{triangle-median}
\begin{tikzpicture}
  % second finished figure
\end{tikzpicture}
```

25.11.3 Stage 3: Rebuild One Picture

```
\tikzset{external/remake next}  
\tikzsetnextfilename{triangle-median}  
\begin{tikzpicture}  
  % revised triangle figure  
\end{tikzpicture}
```

The graph remains cached while the revised triangle is rebuilt.

25.12 A Production Checklist

Before a release build:

1. compile every figure normally during development;
2. stabilize project-wide styles and component definitions;
3. assign descriptive, unique externalized names;
4. rebuild after shared dependencies change;
5. make one clean production build; and
6. visually inspect the resulting document.

Externalization improves speed, but it does not replace visual verification.

25.13 Debugging Workshop

25.13.1 No External Files Appear

Check that the library is loaded, externalization is activated, the compilation command permits the required process, and the cache directory is writable.

25.13.2 A Picture Uses Old Styling

Rebuild that picture or the affected cache after changing shared definitions.

25.13.3 The Wrong Picture Receives a Name

Place `\tikzsetnextfilename` immediately before its intended `tikzpicture`.

25.13.4 Only One Picture Fails

Disable externalization temporarily and compile that picture normally. Fix the TikZ source first, then re-enable the production workflow.

25.14 Exercises

1. Add the minimal externalization setup to a two-picture document.
2. Direct generated files to a cache directory.
3. Give both pictures stable semantic names.
4. Exclude one inexpensive picture.
5. Request a rebuild of only the next picture.
6. Explain when a broad rebuild is necessary.
7. Separate authored figure source from generated artifacts.
8. Write a clean production-build checklist for a book.

25.15 Solutions

25.15.1 Exercises 1–4

```
\usetikzlibrary{external}  
\tikzexternalize[prefix=tikz-cache/]  
\tikzsetnextfilename{first-semantic-name}  
% first tikzpicture  
\tikzset{external/export next=false}  
% inexpensive tikzpicture
```

25.15.2 Exercises 5–7

```
\tikzset{external/remake next}  
\tikzsetnextfilename{picture-to-rebuild}
```

Exercise 6 uses a broad rebuild after a shared style or component changes. Exercise 7 keeps editable source outside the generated cache.

For Exercise 8, end with a clean rebuild and visual inspection of every changed or expensive figure.

25.16 ChatGPT Workshop

Ask ChatGPT to diagnose production configuration without re-designing the figures.

1. Add the smallest safe externalization setup to this existing document.
2. Propose unique semantic cache names for these figures.
3. Diagnose why the cache directory remains empty.
4. Decide whether one picture or the full cache must be rebuilt.
5. Separate authored files, generated files, and release-build checks.

25.17 Final Checklist

Before continuing to Chapter 26, check that you can

1. decide whether externalization is worthwhile;
2. activate it with an appropriate compilation command;
3. isolate generated files in a cache directory;
4. assign stable semantic names;
5. exclude unsuitable pictures;
6. choose a selective or broad rebuild; and
7. complete a clean, visually checked production build.